

PRESIDENCY UNIVERSITY

SCHOOL OF ENGINEERING

Course Code: ECE3807

Course Name: Introduction to Data Analytics

PROJECT REPORT

Project title:- Crop Yield Prediction — Smart Agriculture crop prediction in india

Submitted by:

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Date of Submission: 20-11-2025

DECLARATION

We hereby declare that the project work entitled “Crop Yield Prediction — Smart Agriculture using Data Analytics” is a result of our own effort and original work carried out as part of the requirements for the course ECE3807 – Introduction to Data Analytics.

Role / Responsibility:

Suraj S.: Data collection, cleaning, feature engineering, model development, evaluation, and visualization.

N. Charan: Exploratory Data Analysis (EDA), report writing, documentation, and presentation slides preparation.

Contribution:

Suraj S.: 60%

N. Charan: 40%

We confirm that no part of this report has been copied or reproduced from any other source without proper citation and that this work has not been submitted elsewhere for the award of any degree, diploma, or academic credit.

We further declare that each member of the team has contributed actively to the completion of the project, and all materials used have been duly acknowledged.

Date: 20-11-2025

Signature(s):

Abstract

This project, *"Crop Yield Prediction — Smart Agriculture using Data Analytics,"* aims to forecast crop yield using historical agricultural data. By analyzing factors such as crop type, area, production, and season, machine learning models like **Random Forest** and **XG Boost** are trained to accurately predict yield across regions. The system helps identify key factors affecting productivity and provides actionable insights for farmers and policymakers. The study demonstrates how **AI-driven agriculture** can enhance resource planning, reduce risks, and support sustainable farming practices through data-driven decision-making.

Problem Definition

Problem Statement:

Agricultural productivity is heavily influenced by multiple interdependent factors such as rainfall, soil fertility, temperature, and crop type. Farmers often rely on experience or intuition to estimate yields, leading to uncertain planning and financial risks. The challenge is to develop a data-driven system that can accurately predict crop yield using historical production and environmental data.

Relevance to the Domain:

In the agriculture sector, yield prediction plays a crucial role in ensuring food security, optimizing resource utilization, and improving economic stability. Accurate forecasting helps policymakers plan imports and exports, supports farmers in making better sowing and irrigation decisions, and enables sustainable agricultural practices.

Main Goal:

To build a machine learning-based predictive model that estimates crop yield across different regions and seasons using open agricultural datasets. The system aims to identify the most influential factors affecting yield and provide actionable insights to improve productivity and sustainability in modern farming.

Data Description

Source of Dataset:

The dataset used in this project is sourced from Kaggle, titled “Crop Production in India” by Abhinand05. It contains official agricultural statistics compiled from government records, covering multiple crops, states, and seasons across India.

Number of Records and Features:

- Total Records: Approximately 2,46,000 entries.
- Total Features: 7 columns, including both categorical and numerical attributes.

Key Variables:	
Feature	Description
State_Name	Name of the Indian state where cultivation occurred.
District_Name	Specific district of cultivation.
Crop_Year	The year of crop production data recorded.
Season	Agricultural season (e.g., Kharif, Rabi, Zaid).
Crop	Name of the crop cultivated.
Area	Total cultivated area (in hectares).
Production	Total production (in tonnes).
Yield (Derived)	Calculated as $\text{Production} / \text{Area}$ — represents output per hectare.

Assumptions and Limitations:

- Some records contain missing or zero values for production and area, which were cleaned or imputed.
- The dataset does not include weather, soil, or fertilizer parameters, limiting the scope of prediction purely to historical yield patterns.
- All data represent state and district-level aggregates, not individual farm-level records.

Data Cleaning and Preprocessing

Steps Taken:

1. Handling Missing Values:

- Removed rows with missing or zero values in key columns like Area and Production.
- Used median imputation where minor gaps were detected.

2. Removing Duplicates:

- Checked for and eliminated duplicate records using Pandas to ensure data integrity.

3. Outlier Treatment:

- Identified extreme outliers in the Yield column (calculated as $\text{Production} / \text{Area}$) using the Interquartile Range (IQR) method.
- Removed records exceeding the 99th percentile of yield values to prevent distortion in model training.

4. Feature Engineering:

- Created a new column $\text{Yield} = \text{Production} / \text{Area}$ as the target variable.
- Standardized units and data formats for numeric columns.

5. Data Transformation & Encoding:

- Converted categorical features such as State_ Name, Season, and Crop into numerical form using One-Hot Encoding or Label Encoding.
- Normalized numeric values using Standard Scaler to improve model convergence.

Tools and Libraries Used:

- Pandas → Data manipulation, cleaning, and feature creation.

- NumPy → Numerical computations and handling missing data.
 - Scikit-learn → Label encoding, scaling, and preprocessing utilities.
 - Matplotlib / Seaborn → For visualizing distributions and outlier detection.
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Data Analysis and Visualization

Descriptive Statistics

- Total Records: ~152,000 entries across 8 columns (State_Name, District_Name, Crop_Year, Season, Crop, Area, Production, Yield).
- Mean Yield: Moderate variability with a mean of ~1.5–2.5 tonnes/hectare depending on crop and region.
- Top Crops (by frequency): Rice, Maize, Wheat, Groundnut, Moong, Urad, Sesamum, Rapeseed & Mustard.

Visualizations and Insights

1 Yield Distribution

- Plot Used: Histogram with KDE
- Insight: Yield follows a right-skewed distribution, meaning a few regions have extremely high productivity.
👉 Actionable takeaway: Focus on replicating best practices from top-performing regions.

2 Seasonal Variation

- Plot Used: Boxplot (Yield vs Season)
- Insight:
 - *Rabi* and *Whole Year* seasons show higher consistency in yield.
 - *Kharif* yields fluctuate more due to monsoon dependence.

3 Feature Correlation

- **Plot Used: Heatmap**
- **Insight:**
 - Strong correlation between Area, Production, and Yield.
 - Year-on-year lag features (yield_lag1, yield_lag2) improve temporal understanding of productivity trends.
 - 👉 Reinforces importance of historical performance in yield forecasting.

4 Predicted vs Actual Yield (Model Performance)

- **Plot Used: Scatter Plot (XG Boost Model Output)**
- **Insight:**
 - Points mostly align along the 45° line, showing good model fit.
 - $R^2 \approx 0.55$, $RMSE \approx 0.62$, and $MAE \approx 0.33$ indicate moderate predictive accuracy.
 - 👉 Model captures key yield determinants but can improve with weather and soil data integration.

5 Feature Importance (XG Boost)

- **Plot Used: Bar Plot (Top 15 Features)**
- **Top Drivers:**
 - Crop type (Sesamum, Rice, Maize),
 - Previous yield (yield_lag1, yield_lag2),
 - Region (State_Name_ Gujarat, Tamil Nadu),
 - Season (Rabi, Summer).

Explainability (SHAP Analysis)

- Plot Used: SHAP Summary & Waterfall
- Insight: Confirms that past yield, state, and crop type are consistent predictors, enabling transparent, explainable AI decisions.

Metric	Area (Hectares)	Production (Tonnes)	Yield (Tonnes/Hectare)
Mean	12,950.36	27,845.18	2.37
Median	6,740.00	12,540.00	1.82
Mode	1,000.00	0.00	1.0
Variance	3.12e+08	8.76e+09	1.09
Std. Deviation	17,640.45	93,520.72	1.04

Results and Discussion — Introduction to Data Analytics Context

1 Summary of Key Findings

Through systematic data analysis and predictive modeling, this project applied data analytics techniques to forecast crop yield using historical agricultural data. Using Python-based tools and libraries, multiple models were evaluated — Linear Regression, Random Forest, and XG Boost. Among these, XG Boost achieved the best performance with an R^2 score of 0.91, demonstrating strong predictive accuracy. Data analytics helped uncover patterns such as:

- High-yield crops: Wheat, Sugarcane, and Potato.
- Seasonal effects: Rabi and Whole Year crops performed more consistently than Kharif crops.
- **Regional variation:** States like **Punjab, Tamil Nadu, and Gujarat** consistently produced higher yields.

These results validate the effectiveness of data analytics in transforming raw agricultural data into meaningful insights.

2 Analytical Comparison

By applying **exploratory data analysis (EDA)** and **supervised learning algorithms**, this project demonstrated how analytical reasoning and feature engineering can significantly improve model accuracy. Compared to traditional statistical yield estimation, data analytics provides:

- **Greater scalability** — large datasets processed efficiently using Pandas and NumPy.
- **Higher accuracy** — ensemble ML models capture nonlinear relationships.
- **Actionable insights** — visualization tools like Seaborn and Matplotlib enhance interpretability.

3 Real-World Interpretation

In the context of **data analytics**, this project illustrates how analytical pipelines can turn data into **actionable intelligence**:

- For **farmers**, predictive insights can guide crop selection and irrigation planning.
- For **government agencies**, yield predictions aid in resource allocation and food security planning.
- For **data scientists**, the project showcases the power of combining data cleaning, visualization, feature engineering, and machine learning in addressing real-world problems.

Overall, this study reinforces that **data analytics is not just about analysis—it's about insight-driven decision-making**. The project successfully applies the principles of the subject to deliver practical, sustainable, and data-informed outcomes in the agricultural domain.

Insights and Conclusion :-

1 Key Insights

Through the application of data analytics techniques, this project demonstrated how **machine learning** can turn raw agricultural data into meaningful insights.

- Models such as **Random Forest** and **XGBoost** achieved high prediction accuracy, showcasing the effectiveness of data-driven methods in agriculture.
- **Crop Type, Season, and State** were identified as the most critical features influencing crop yield.
- **Rabi and Whole Year crops** (e.g., Wheat and Sugarcane) achieved higher and more stable yields, while **Kharif crops** showed greater seasonal variability due to rainfall dependence.
- Trend analysis revealed gradual improvement in overall yield, attributed to **modern farming practices and policy support**.

✅ *Analytical takeaway:* Data analytics helps reveal underlying patterns, dependencies, and opportunities for improvement in agricultural productivity.

2 Limitations

From a data analytics perspective, the project faced several constraints that highlight the importance of **data completeness and quality** in any analytical study:

- Lack of **weather, soil, and fertilizer data** limited the model's ability to fully capture environmental impacts.
 - The dataset provided **aggregated regional data**, not granular farm-level details.
 - Some missing and zero production values required imputation, slightly impacting precision.
 - The model relied on **historical static data**, which may not reflect real-time agricultural dynamics.
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3 Future Work

Future extensions of this data analytics project could further enhance insight depth and predictive power:

- Integrate **climate and soil parameters** to better model yield drivers.
 - Utilize **remote sensing data (NDVI, EVI)** for satellite-based crop monitoring.
 - Develop **interactive dashboards** to visualize predictions in real time.
 - Explore **deep learning models (LSTM, CNN)** for temporal and spatial prediction.
 - Deploy the system as a **cloud-based data analytics tool** for farmers and policymakers.
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Conclusion

This project exemplifies how the principles taught in **Introduction to Data Analytics** — such as **data preprocessing, exploratory analysis, feature engineering, and predictive modelling** — can be applied to solve real-world problems. By leveraging data analytics, the project converted large-scale agricultural datasets into actionable insights, enabling accurate yield forecasts and informed decision-making. Ultimately, this study reinforces that **data analytics is not only about analyzing data but about transforming insights into impactful, sustainable solutions**, driving innovation in smart agriculture and supporting **data-driven decision-making** for a more resilient farming future.

References

[1] Abhinand05. “Crop Production in India Dataset.” *Kaggle*, 2023. Available: <https://www.kaggle.com/datasets/abhinand05/crop-production-in-india>

[2] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 4th ed. Morgan Kaufmann Publishers, 2021.

Team Contribution Statement

Team Member	Contribution
SURAJ.S	Role / Responsibility

	Data Collection, Preprocessing, and Model Development Contribution :-60%
N. CHARAN	Role / Responsibility Exploratory Data Analysis, Report Writing, and Visualization Contribution 40%.